MINOR PROJECT

Diabetes Prediction

Logistic regression on the Pima Indians diabetes dataset

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Objective and dataset

The goal is to classify Outcome (0 = no diabetes, 1 = diabetes) from eight health-related measurements. The supplied CSV contains 768 records. Zero values in Glucose, BloodPressure, SkinThickness, Insulin, and BMI were treated as missing measurements.

Workflow: stratified 80/20 train-test split (random_state=42), median imputation, StandardScaler, then class-balanced Logistic Regression. The imputer is fitted on training data only.

Inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

Holdout results

ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-AUC
73.4%	60.3%	70.4%	65.0%	0.813

Confusion matrix

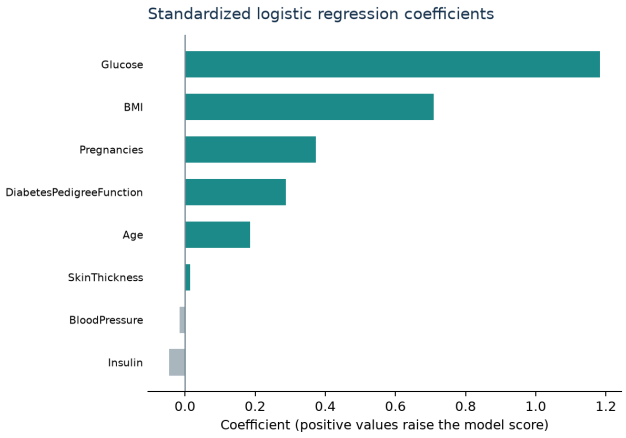
Actual rows | Predicted columns

	PREDICTED 0	PREDICTED 1
ACTUAL 0	75	25
ACTUAL 1	16	38

Model workflow

- Invalid zero measurements marked as missing
- Stratified split preserves outcome proportions
- Median imputation and scaling fitted on training rows
- Balanced logistic regression with default 0.50 threshold

What the model learned



Glucose had the largest positive standardized coefficient (1.18); **BMI** was next (0.71). These are associations learned by the model, not evidence of cause.

Recall was 0.704: the model found 38 of 54 positive test cases. It missed 16, so recall is useful alongside accuracy.

Coefficient sizes are on standardized inputs. They describe how this fitted model changes its score and are not clinical risk estimates.

CONCLUSION

On this fixed split, the model reached a ROC-AUC of 0.813 and detected about 70.4% of positive cases. This small historical dataset represents a specific population. The analysis is for coursework only and must not be used to diagnose or guide care.

Source: supplied diabetes.csv (Pima Indians Diabetes Database). Metrics are from one stratified holdout split.

Educational analysis only. It is not a medical diagnostic tool.